

ECONOMICS & APPLIED STATISTICS

Official Ministry of Education Textbook - Grade 12

ACADEMIC YEAR: 2025 / 2026 | GRADE 12 (GENERAL SECONDARY CERTIFICATE)

CURRICULUM OVERVIEW & AUTHORIZED SCOPE

The official Ministry of Education textbook for Grade 12 Economics and Applied Statistics, presenting rigorous coverage of scarcity, market mechanisms, macroeconomic income loops, public budgeting, monetary policy, and foundational applied statistics (Pearson, Spearman, Bayes theorem, and standard Gaussian distributions).

CORE CURRICULUM HIGHLIGHTS:

- Microeconomic scarcity, opportunity cost, and production possibility boundaries
- Keynesian national income accounting: Output, Expenditure, and Value-Added streams
- Fiscal budgeting mechanisms, state debt, tax classification, and market failures
- Monetary aggregates (M1, M2), central banking reserve ratios, and credit creation
- Pearson product-moment and Spearman rank correlation coefficients with line of best fit
- Conditional probability axioms, Bayes tree diagrams, and mutual independence criteria
- Discrete and continuous probability density functions with expected value and variance
- Gaussian bell curve transformations to standard Z-scores and empirical probability tables

APPROVED CHAPTER MODULES:

- Ch 1: The Economic Problem & Scarcity	pp. 1-24 (8 core units)
- Ch 2: Factors of Production & Capital Formation	pp. 25-48 (8 core units)
- Ch 3: National Income & Economic Growth	pp. 49-74 (8 core units)
- Ch 4: Public Finance & Fiscal Policy	pp. 75-100 (8 core units)
- Ch 5: Money, Banking & Monetary Systems	pp. 101-126 (8 core units)
- Ch 6: International Economic Relations & Globalization	pp. 127-152 (8 core units)
- Ch 7: Linear Correlation & Regression Analysis	pp. 153-174 (8 core units)
- Ch 8: Conditional Probability & Independent Events	pp. 175-196 (8 core units)

OFFICIAL PUBLICATION METADATA & VERIFICATION

Document: Official Economics & Applied Statistics Formula Compendium (38p - 3.8 MB)
Full Ministerial Textbook: 240 Pages - 26.8 MB (Direct on MOE Portal)
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Official Portal: <https://moe.gov.eg/ar/elearning-content/> - Code: TH-ECON-STAT-G12
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MINISTERIAL PREFACE & STATUTORY FRAMEWORK

MINISTERIAL DECREE NO. 138 OF 2025

Pursuant to the decisions of the Supreme Council for Pre-University Education, this textbook is officially accredited as the national core curriculum for Senior Secondary and Baccalaureate qualifications for the 2025/2026 academic cycle.

1. Pedagogical Scope & National Objectives

The official Ministry of Education textbook for Grade 12 Economics and Applied Statistics, presenting rigorous coverage of scarcity, market mechanisms, macroeconomic income loops, public budgeting, monetary policy, and foundational applied statistics (Pearson, Spearman, Bayes theorem, and standard Gaussian distributions). Microeconomic scarcity, opportunity cost, and production possibility boundaries Keynesian national income accounting: Output, Expenditure, and Value-Added streams Fiscal budgeting mechanisms, state debt, tax classification, and market failures Monetary aggregates (M1, M2), central banking reserve ratios, and credit creation Pearson product-moment and Spearman rank correlation coefficients with line of best fit Conditional probability axioms, Bayes tree diagrams, and mutual independence criteria Discrete and continuous probability density functions with expected value and variance Gaussian bell curve transformations to standard Z-scores and empirical probability tables

2. Assessment Cognitive Taxonomy Specification

All ministerial evaluation instruments conform strictly to the following weight distribution:

30% Knowledge & Recall Definitions, laws & formulas	45% Understanding & Application Calculations, proofs & models	25% Higher Order Thinking (HOTS) Analysis, synthesis & non-routine
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3. Approved Scientific Calculators & Examination Instrumentation

- Approved Calculators: Non-programmable scientific calculators (Casio fx-991ARX, fx-991CW, or equivalent approved models).
- Prohibited Devices: Any device with symbolic CAS (Computer Algebra System), programmable memory, internet connectivity, or graphic display capabilities is strictly prohibited in examination centers.
- Formula Reference Sheets: The official ministerial formula sheet is printed directly in the examination booklet; separate personal scratch notes are forbidden.
- Drawing Instruments: Standard geometric compass, protractor, 30/60 and 45 degree set squares, and transparent rulers are mandatory.

4. Egyptian Knowledge Bank (EKB) & Digital Portal Synchronisation

Students and teachers can access interactive simulations, GeoGebra 3D interactive graphs, video lectures by Senior Curriculum Inspectors, and self-assessment question banks at the official Ministry of Education digital portal:

<https://moe.gov.eg/ar/elearning-content/>. Scanning the QR code on the back cover links directly to this textbook's digital resource pack.

SYLLABUS BLUEPRINT & EXAMINATION WEIGHT MATRIX

Official Duration: 180 Minutes (3 Hours) | Total Paper Marks: 60 Marks

Chapter / Modular Title	Weeks	Marks	Weight	Curricular Focus
Ch 1: The Economic Problem & Scarcity	3 wks	6 pts	10.0%	8 core instructional units covering national standards and
Ch 2: Factors of Production & Capital Formation	3 wks	6 pts	10.0%	8 core instructional units covering national standards and
Ch 3: National Income & Economic Growth	3 wks	6 pts	10.0%	8 core instructional units covering national standards and
Ch 4: Public Finance & Fiscal Policy	3 wks	6 pts	10.0%	8 core instructional units covering national standards and
Ch 5: Money, Banking & Monetary Systems	3 wks	6 pts	10.0%	8 core instructional units covering national standards and
Ch 6: International Economic Relations & Globalization	3 wks	6 pts	10.0%	8 core instructional units covering national standards and
Ch 7: Linear Correlation & Regression Analysis	3 wks	6 pts	10.0%	8 core instructional units covering national standards and
Ch 8: Conditional Probability & Independent Events	3 wks	6 pts	10.0%	8 core instructional units covering national standards and
Ch 9: Random Variables & Probability Distributions	3 wks	6 pts	10.0%	8 core instructional units covering national standards and
Ch 10: Standard Normal Distribution & Z-Scores	3 wks	6 pts	10.0%	8 core instructional units covering national standards and

INTERDISCIPLINARY COGNITIVE BRIDGES

Physics (Mechanics & Electromagnetism) Direct mathematical articulation with rotational kinetics, projectile kinematics, electric flux integrations, and harmonic oscillators.
Engineering Sciences (Civil, Mechanical, Electrical) Matrix state-space modeling, 3D spatial stress analysis, moments of inertia, and equilibrium structural design.
Computer Science & Algorithmics Discrete combinatorics, graph search trees, Boolean logic transformations, algorithmic asymptotic complexity, and linear transformations.

CHAPTER SPECIFICATION: CH 1: THE ECONOMIC PROBLEM & SCARCITY

A. Targeted Ministerial Learning Outcomes (Cognitive Standards)

- [+] Analyze fundamental principles, statutory definitions, and theoretical models of Ch 1: The Economic Problem & Scarcity.
- [+] Synthesize and apply analytical frameworks across all 8 modular curriculum units.
- [+] Demonstrate mastery of standardized ministerial examination competencies.
- [+] Solve higher-order analytical problems and contextual application scenarios.

B. Statutory Theorems & Analytical Principles

Ministerial Framework Standard 1.1: Primary foundational doctrines and axiomatic structures of Ch 1: The Economic Problem & Scarcity.

Curricular Analytical Directive 1.2: Standard methodological paradigms and diagnostic criteria.

C. Standard Mathematical Expressions & Operational Formulas

- > Operational Assessment Metric: $\Phi_1 = \frac{\sum (\omega_k * \psi_k)}{\sum \omega_k}$
- > Curricular Index Specification: $\Lambda(x) = \alpha * f(x) + \beta * g(x)$
- > Standardized Ministerial Model: $\Delta = \sqrt{\frac{1}{N} * \sum_{i=1}^N (x_i - \bar{x})^2}$

D. National Examination Pitfalls & Common Student Errors

[!] Overlooking official ministerial terminology and structured proof/solution steps.

[!] Confusing related foundational concepts under timed assessment conditions.

[!] Failing to provide complete reasoning and units for operational answers.

CHAPTER SPECIFICATION: CH 2: FACTORS OF PRODUCTION & CAPITAL FORMATION

A. Targeted Ministerial Learning Outcomes (Cognitive Standards)

- [+] Analyze fundamental principles, statutory definitions, and theoretical models of Ch 2: Factors of Production & Capital Formation.
- [+] Synthesize and apply analytical frameworks across all 8 modular curriculum units.
- [+] Demonstrate mastery of standardized ministerial examination competencies.
- [+] Solve higher-order analytical problems and contextual application scenarios.

B. Statutory Theorems & Analytical Principles

Ministerial Framework Standard 2.1: Primary foundational doctrines and axiomatic structures of Ch 2: Factors of Production & Capital Formation.

Curricular Analytical Directive 2.2: Standard methodological paradigms and diagnostic criteria.

C. Standard Mathematical Expressions & Operational Formulas

- > Operational Assessment Metric: $\Phi_2 = \frac{\sum (\omega_k * \psi_k)}{\sum \omega_k}$
- > Curricular Index Specification: $\Lambda(x) = \alpha * f(x) + \beta * g(x)$
- > Standardized Ministerial Model: $\Delta = \sqrt{\frac{1}{N} * \sum_{i=1}^N (x_i - \bar{x})^2}$

D. National Examination Pitfalls & Common Student Errors

[!] Overlooking official ministerial terminology and structured proof/solution steps.

[!] Confusing related foundational concepts under timed assessment conditions.

[!] Failing to provide complete reasoning and units for operational answers.

CHAPTER SPECIFICATION: CH 3: NATIONAL INCOME & ECONOMIC GROWTH

A. Targeted Ministerial Learning Outcomes (Cognitive Standards)

- [+] Analyze fundamental principles, statutory definitions, and theoretical models of Ch 3: National Income & Economic Growth.
- [+] Synthesize and apply analytical frameworks across all 8 modular curriculum units.
- [+] Demonstrate mastery of standardized ministerial examination competencies.
- [+] Solve higher-order analytical problems and contextual application scenarios.

B. Statutory Theorems & Analytical Principles

Ministerial Framework Standard 3.1: Primary foundational doctrines and axiomatic structures of Ch 3: National Income & Economic Growth.

Curricular Analytical Directive 3.2: Standard methodological paradigms and diagnostic criteria.

C. Standard Mathematical Expressions & Operational Formulas

- > Operational Assessment Metric: $\Phi_3 = \frac{\sum (\omega_k * \psi_k)}{\sum \omega_k}$
- > Curricular Index Specification: $\Lambda(x) = \alpha * f(x) + \beta * g(x)$
- > Standardized Ministerial Model: $\Delta = \sqrt{\frac{1}{N} * \sum_{i=1}^N (x_i - \bar{x})^2}$

D. National Examination Pitfalls & Common Student Errors

[!] Overlooking official ministerial terminology and structured proof/solution steps.

[!] Confusing related foundational concepts under timed assessment conditions.

[!] Failing to provide complete reasoning and units for operational answers.

OFFICIAL MINISTRY FORMULA REFERENCE SHEET

AUTHORIZED CONCEPT CARD (SHEET OF CONCEPTS) PROVIDED IN EXAMINATION CENTERS

ECONOMICS & APPLIED STATISTICS - CORE PRINCIPLES & AXIOMS	
Fundamental Identity:	$\Psi(x) = \sum_{k=1}^n c_k \cdot \phi_k(x)$
Equilibrium Balance Criterion:	$\sum F_{\text{net}} = 0, \sum M_{\text{pivot}} = 0$
Continuity & Conservation Standard:	$\text{div}(\mathbf{J}) + \partial \rho / \partial t = 0$
Normalized Standard Score:	$Z = (X - \mu) / \sigma$

PROCEDURAL ASSESSMENT & TRANSFORMATION RULES	
Linear Operator Transformation:	$T(\alpha \cdot u + \beta \cdot v) = \alpha T(u) + \beta T(v)$
Harmonic Synthesis Mean:	$H = n / (\sum_{i=1}^n 1/x_i)$
Sample Variance Boundary:	$s^2 = (1 / (n - 1)) \cdot \sum (x_i - \bar{x})^2$
Curricular Efficiency Ratio:	$\eta = (\text{Output}_{\text{actual}} / \text{Input}_{\text{standard}}) \cdot 100\%$

NATIONAL EXAMINATION PROTOCOLS & EVALUATION MATRIX

1. Structure of the National Examination Paper

Paper Duration:	180 Minutes (3 Hours)
Total Scaled Marks:	60 Marks
Section I (Objective):	Multiple Choice Questions (MCQ) with 4 options; single best answer.
Section II (Subjective):	Structured multi-step analytical and proof essay questions.
Electronic Scoring:	Objective answer sheets processed via computerized optical mark recognition (OMR).
Manual Essay Scoring:	Double-blind evaluation by two independent certified ministerial examiners.

2. Standard Candidate Exam Directives

- All examination responses must strictly adhere to official ministerial syllabus specifications.
- Clear intermediate analytical steps, definitions, and proper notation are required for full credit.
- Calculators and reference sheets must strictly comply with official Ministry regulations.
- Review the certified digital textbook compendium for complete model solutions and rubrics.

ARAB REPUBLIC OF EGYPT - MINISTRY OF EDUCATION
NATIONAL CENTER FOR EXAMINATIONS & EDUCATIONAL EVALUATION (NCEEE)

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